

28.2 A 400-to-1000nm 24 μ W Monolithic PPG Sensor with 0.3A/W Spectral Responsivity for Miniature Wearables

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Incorporating different wavelength (400 to 1000nm) LEDs, photoplethysmography (PPG) sensors allow wearable devices to monitor various health parameters such as heart rate (HR), oxygen saturation (SpO₂), and blood pressure (BP). Nowadays, PPG sensing technology at the wrist is well established. To cope with the large degree of motion turbulence presented at the wrist, PPG sensors use Green (Gr) LEDs together with multiple photodiodes (PD), and they are driven by wide-dynamic-range (DR) current-sensing front-ends [1]. It is attractive to use a near-infra-red (nIR) PPG sensor in a True Wireless Stereo (TWS), as the ear provides the best site to measure heart rhythm (more blood flow, constant distance from the heart, and less motion than at the finger or wrist). However, TWS requires a PPG sensor that is more stringent on size and power consumption (shown in Fig. 28.2.1). A promising solution [2,3] is integrating an array of PDs with an ADC to dramatically reduce power while also providing monolithic integration. However, the limited DR (<80dB) and the poor spectral responsivity remain challenging. This work advances [1] by demonstrating a CMOS monolithic PPG sensor, and improves spectral responsivity more than 4 $\times$ (0.3A/W across 400 to 1000nm) compared to [2]. The sensor is fabricated by back-side illumination (BSI) CMOS technology providing 90dB DR (18dB improvement from [3]) while consuming only 24 μ W power and 5.5mm² silicon area.

To save power, a PPG sensor preferably operates in a low-light condition with nIR LED (to save LED power) and a PD with high sensitivity at the same wavelength. However, the quantum efficiency (QE) of a conventional CMOS image sensor (CIS) pixel is limited below 10% [2] (corresponds to 0.07A/W at 925nm) because of the thin epitaxial silicon (Si) and absence of high voltage (drift E-field). To improve QE at nIR wavelength, we further optimize several technologies [3] on CIS pixels such as Si thickness (T_{Si}), anti-reflection layer (ARL) thickness, and a back-side scattering (BST) structure. First, Si thickness increases up to 5.1 μ m, such that PDs have sufficient drift E-field with only 1.8V. Second, a dense (1.12 μ m pitch) BST structure is inserted to guide light scattered and reflected repeatedly, which in turn increases the effective T_{Si}. Moreover, the PD (22.4 $\times$ 22.4 μ m²) integrates a full cross BST pattern to optimize QE. A micro lens (ML) further focuses light on the BST structure to improve efficiency of light harvesting. Third, the ARL with thicker dielectric minimizes light loss in the range from 400 to 1000nm. Consequently, the spectral responsivity over 0.3A/W in the broad spectrum (400 to 1000nm) is obtained as shown in Fig. 28.2.2. Compared to a conventional CIS pixel, 0.25A/W improvement at 925nm is achieved (4.6 $\times$ improvement). The light-sensitive area of the on-chip detector is 1.04mm². It uses 4 $\times$ 4 macro pixels where each macro pixel consists of 13 $\times$ 10 PDs merged together. Depending on the user scenario, one can readout each macro pixel sequentially, or in parallel.

As shown in Fig. 28.2.3, a transimpedance amplifier (TIA) supplies photocurrent to the on-chip detector and amplifies the PPG signal into a voltage by a gain of R_{TIA}. To tolerate a large degree ambient light, a 7b current DAC (up to 128 μ A) is added at the input of the TIA for coarse-tuned ambient light cancellation (ALC). Subsequent to the TIA, 2 sample & hold (S&H) filters together with a programmable gain amplifier (PGA) provide: 1) fine-tuned ALC with correlated (in synchronous to LED pulse) double sampling (CDS), 2) low-frequency noise mitigation at the TIA stage, and 3) dynamic range extension by translating the signal from single- to differential-ended. An ADC is integrated to digitize PGA outputs into 16b format, which are saved in a FIFO (depth 64). SPI and I2C are available for the sensor to communicate with an external Microcontroller (MCU). 2-channel LED drivers are integrated, and a timing controller synchronizes them to the receiver (RX) chain. Using a single 1.8V supply, the RX chain consumes 24 μ W on average at a 20Hz sampling rate.

Due to the single-ended nature of the on-chip detector, the RX chain starts with a single input/output TIA biased by a bandgap reference (BGR) and a current DAC (coarse ALC). To mitigate noise ($V_{n,ref}^2$) from the BGR, reference sampling (S_{REF}) before the S&H (S₁ and S₂) is implemented. The PGA cancels out BGR-induced (sampled) noise with the help of the CDS. By controlling the bias current at the output of the PGA, the DC level is restored, adaptive to different gain settings. The ADC, operating in oversampling (1024 $\times$) fashion, further reduces the noise of the PGA stage and quantization level so that the noise floor of entire RX chain is limited by the input stage. Figure 28.2.4 shows the CDS transfer function and SNR of the RX chain. With the help of the 1st-order high-pass filter characteristics of the CDS, the entire RX chain ensures -46dB attenuation below 120Hz. This behavior helps to cancel out BGR noise introduced at the input stage, and results in SNR improvements of the RX chain. As a result, 90dB SNR (11dB improvement) at -1dB input full scale (FS) is achieved by consuming 1.8mA when the RX chain is active. In addition, it also effectively removes (fine-tuned ALC) low-frequency (<120Hz) noise of ambient light by operating the LED driver synchronously to the S&H.

The sensor chip is integrated with 540nm and 925nm LEDs in a tiny module (3.7mm $\times$ 4.9mm $\times$ 0.5mm (thickness)). By measuring the loop-back SNR (Fig. 28.2.5) including the LED driver and entire RX chain, the sensor achieves >88dB SNR up to 50% scale (25mA) of I_{LED}. Note that the degradation (-8dB) of loop-back SNR at full scale of I_{LED} is due to the excessively small R_{LED} in the LED driver. As shown in Fig. 28.2.6, a custom-designed wireless ear set is developed to measure the PPG signal in real time. Thanks to improved responsivity of the detector, nIR-PPG recording shows comparable power efficiency (34% AC @ 57% LED power compared to Green LED) to Gr-PPG at the ear. Figure 28.2.6 also demonstrates PPG-based HR and BP evaluation. The ear set is equipped with a motion-tolerable algorithm [5], and it can differentiate HR with mean absolute difference (MAD) of 1.9bpm under presence of excessive motion. Without motion, the ear set can evaluate BP at the ear by decomposing the nIR-PPG signal precisely. Compared to a standard cuff BP monitor, it shows 0.67 to 0.71 correlation and MAD of 3.50/4.09mmHg at systolic/diastolic cycle, respectively. Notice that these results are comparable to a state-of-the-art wrist-worn BP monitor.

Figure 28.2.7 compares key performance metrics of the PPG sensors. Compared to [1], this work provides a monolithic sensor solution while consuming only 1/3 of the power. Beneficial in terms of improved spectral responsivity (>0.3A/W at 400 to 1000nm) and low-noise (90dB SNR) RX chain, the sensor enables a TWS to record HR continuously but accurately with motion while consuming only 24 μ W. Moreover, BP evaluation is also possible by analyzing nIR-PPG measured at the ear. These achievements are a significant advance over other monolithic solutions [2,3] and pave the way to integrating a monolithic PPG sensor in a miniature wearable.

References:

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- [2] A. Caizzone et al., "A 2.6 μ W Monolithic CMOS Photoplethysmographic Sensor Operating with 2 μ W LED Power," *ISSCC*, pp. 290-291, Feb. 2019.
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- [5] Z. Zhang et al., "Photoplethysmography-based Heart Rate Monitoring in Physical Activities via Joint Sparse Spectrum Reconstruction," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 8, pp. 1902-1910, 2015.

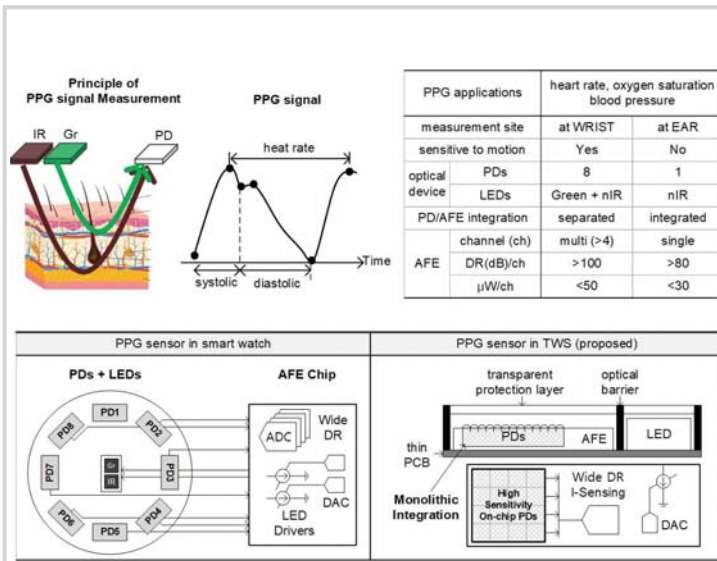


Figure 28.2.1: PPG sensor in wearables.

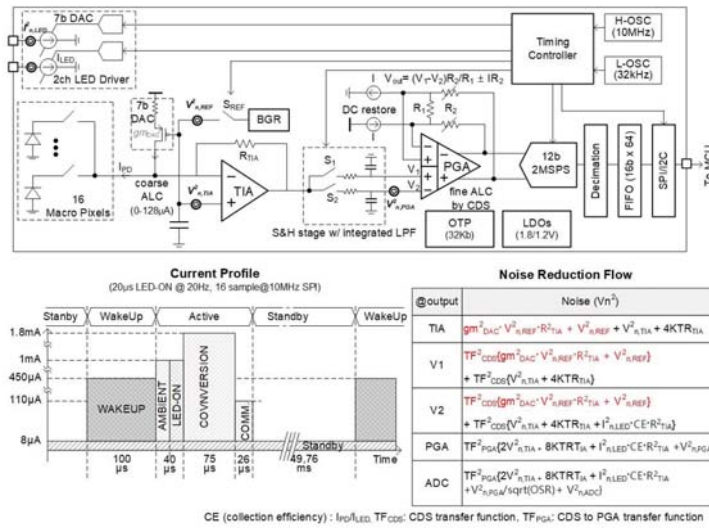


Figure 28.2.3: Architecture of monolithic PPG sensor chip with noise-reduction flow.

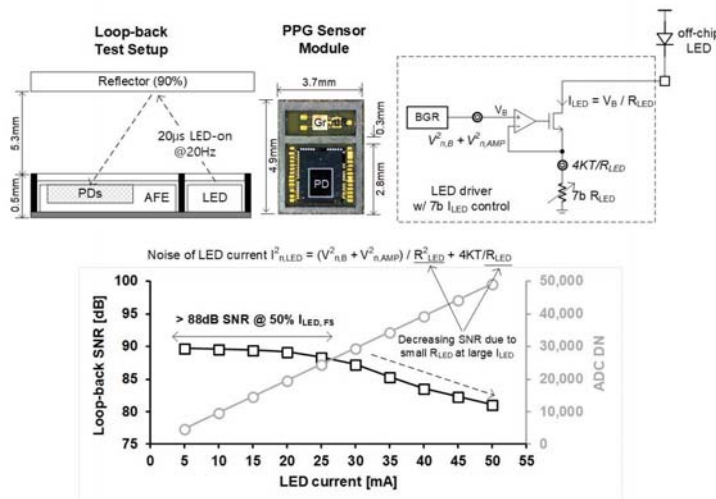


Figure 28.2.5: Loop-back SNR with PPG sensor module.

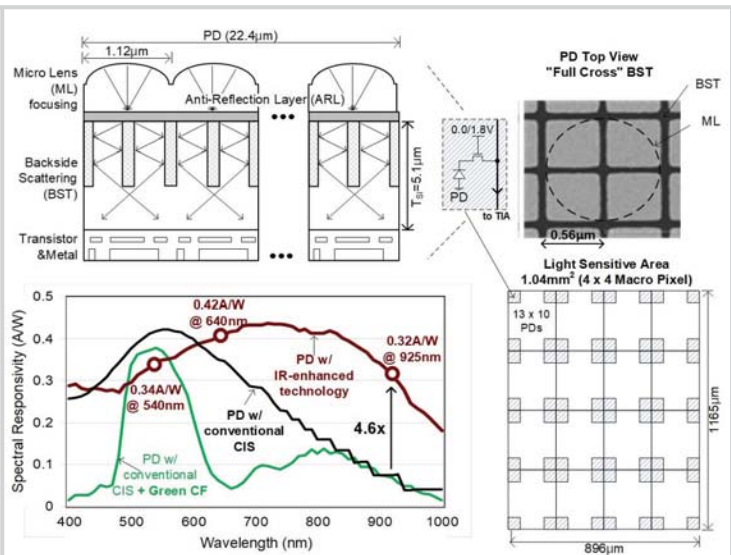


Figure 28.2.2: On-chip detector with an array of sensitivity-enhanced photodiode (PD).

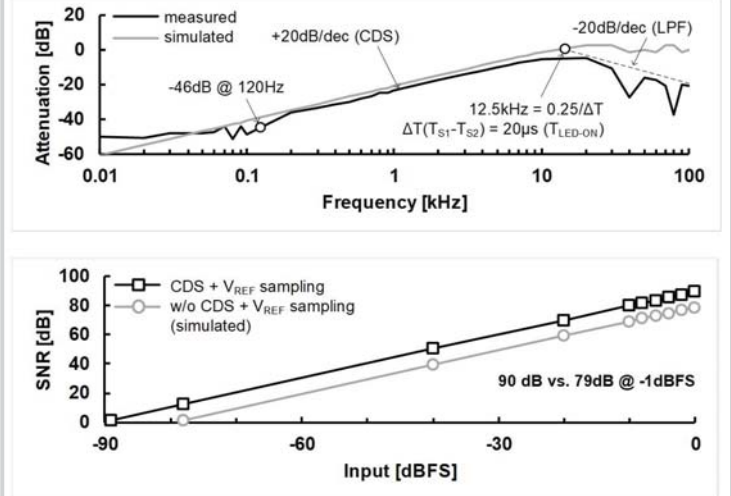


Figure 28.2.4: Measurements of CDS transfer function and SNR in RX chain.

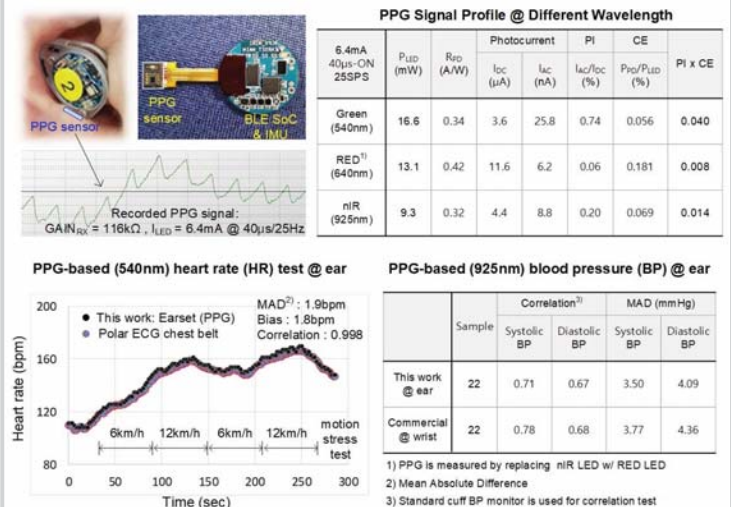


Figure 28.2.6: PPG signal profile from ear and its validation for HR and BP evaluation.

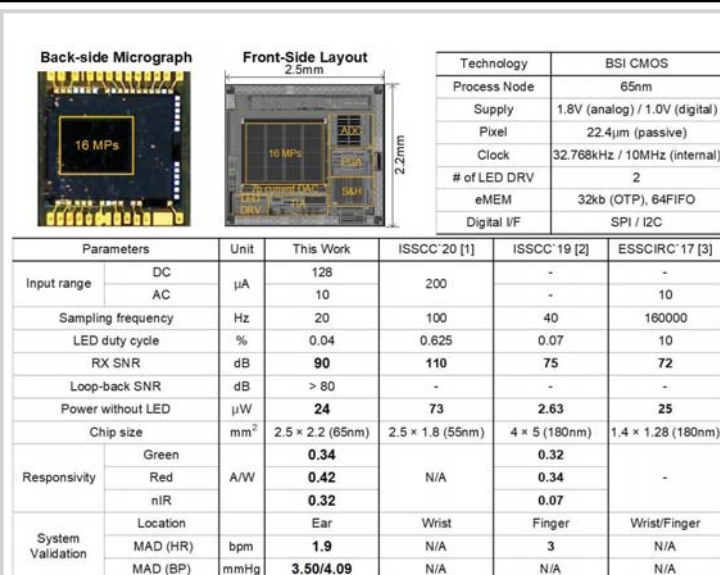


Figure 28.2.7: Performance comparison.